Exercises Control Flow

Exercise

Write a Python program to find whether a given number (accept from the user) is even or odd, print out an appropriate message to the user.

```
num = int(input("Enter a number: "))
mod = num % 2
if mod > 0:
    print("This is an odd number.")
else:
    print("This is an even number.")
```

Exercise

Write a Python program to find those numbers which are divisible by 7 and multiple of 5, between 1500 and 2700 (both included).

```
nl=[]
for x in range(1500, 2701):
    if (x%7==0) and (x%5==0):
        nl.append(str(x))
print (','.join(nl))
```

Exercise

Write a Python program to guess a number between 1 to 9.

Note : User is prompted to enter a guess. If the user guesses wrong then the prompt appears again until the guess is correct, on successful guess, user will get a "Well guessed!" message, and the program will exit.

```
import random
target_num, guess_num = random.randint(1, 10), 0
while target_num != guess_num:
    guess_num = int(input('Guess number between 1 and 10 until you get it : '))
print('Well guessed!')
```

Exercise

Write a Python program to count the number of even and odd numbers from a series of numbers.

```
numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9) # Declaring the tuple
count_odd = 0
count_even = 0
for x in numbers:
```

```
        if not x % 2:
           count_even+=1
        else:
           count_odd+=1
print("Number of even numbers :",count_even)
print("Number of odd numbers :",count_odd)
```

Exercise

Write a Python program that prints all the numbers from 0 to 6 except 3 and 6. Use 'continue' statement.

```
for x in range(6):
    if (x == 3 or x==6):
        continue
    print(x,end=' ')
print("\n")
```

Exercise

Write a Python program to get the Fibonacci series between 0 to 50.

Note : The Fibonacci Sequence is the series of numbers :
0, 1, 1, 2, 3, 5, 8, 13, 21, ...
Every next number is found by adding up the two numbers before it.

```
x,y=0,1

while y<50:
    print(y)
    x,y = y,x+y
```

Exercise

Write a Python program that accepts a string and calculate the number of digits and letters.

```
s = input("Input a string")
d=l=0
for c in s:
    if c.isdigit():
        d=d+1
    elif c.isalpha():
        l=l+1
    else:
        pass
print("Letters", l)
print("Digits", d)
```

Exercise

Write a Python program to check whether an alphabet is a vowel or consonant.

```
l = input("Input a letter of the alphabet: ")

if l in ('a', 'e', 'i', 'o', 'u'):
      print("%s is a vowel." % l)
elif l == 'y':
      print("Sometimes letter y stand for vowel, sometimes stand for
consonant.")
else:
      print("%s is a consonant." % l)
```

Exercise

Write a Python program to convert month name to a number of days.

```
print("List of months: January, February, March, April, May, June, July,
August, September, October, November, December")
month_name = input("Input the name of Month: ")

if month_name == "February":
      print("No. of days: 28/29 days")
elif month_name in ("April", "June", "September", "November"):
      print("No. of days: 30 days")
elif month_name in ("January", "March", "May", "July", "August", "October",
"December"):
      print("No. of days: 31 day")
else:
      print("Wrong month name")
```